Pranav Doma

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EDUCATION

New York University | MS, Computer Engineering | CGPA - 3.73 08/2022 - 05/2024
Coursework: Robot Perception, Computing Systems Architecture, Real-Time Embedded Systems, Deep Learning, Digital Signal Processing, Machine Learning, Applications of AI for Predictive Maintenance (NVIDIA DLI)
S.V. National Institute of Technology | B.Tech, Electrical Engineering | CGPA - 7.24 07/2017 - 06/2021

CERTIFICATIONS

Certified Operator - Autonomous Inspection (Boston Dynamics Spot, on-site training) 10/2026
Certificate Number LP625, valid for 1 year

SKILLS

C, C++, Embedded C, Python, CUDA, MATLAB/Simulink, ROS, Iceoryx/Iceoryx2, Git, GitHub Actions, MuJoCo (MJX), Brax, OpenCV, PyTorch, ONNX, CARLA, Omniverse, Isaac Sim/Lab, Motion Planning, Sensor Fusion, Visual SLAM, Embedded Systems for Robotics, Oracle Cloud, Linux, Docker, Podman, HPC (Slurm, Singularity), Atlassian, CrewAI, Groq, LangChain.

WORK EXPERIENCE

HCLTech America 02/2026 - Present
Robotics Application Engineer (Autonomous Vehicles & Robotics) | *New Brunswick*

- Coauthored SimReadyBench and IWIBench, open benchmarks for simulation ready asset generation and gradient free model evaluation.
- Led the transition from traditional modular pipelines to end-to-end (E2E) autonomous driving architectures, developing auto-labeling algorithms that use existing camera and radar perception models to generate supervision signals for training.
- Developed and released VisionPilot (all versions), an open-source Level 2 ADAS system for automotive OEMs that provides real-time 3D world understanding on edge devices from a single front-facing camera. Built and integrated the longitudinal and lateral network fusion and state-estimation modules using Kalman/particle filtering, and ran system-level testing across SODA Sim, CARLA, and AlpaSim, improving high-speed and sharp-turn driving performance.
- Built QA, inspection, and vision-based testing demos within HCLTech's Kinetic AI platform, including mobile phone QA testing with Vision-Language Models (VLMs), smart-glasses validation, visual inspection workflows, and robotic test automation using UR robots, LeRobot, Trossen WidowX AI, and Unitree Go2 for inspection.
- Developed multimodal robotic QA demonstrations using ACT, VLA-JEPA, and NVIDIA GR00T 1.7, with RGB-D perception, multi-camera data collection, and manipulation workflows, showing autonomous testing, inspection, and validation under the Kinetic QA framework.
- Built an end-to-end Physical AI pipeline on NVIDIA's open Alpa ecosystem, converting non-NVIDIA autonomous-driving data into simulation-ready environments via NCore/NuRec, integrating Fixer-enhanced novel-view rendering, and running native Alpamayo closed-loop driving with reasoning-overlay visualization in AlpaSim.
- Developed a cross-platform AV demonstration framework combining reconstructed 3D scenes, VLM-based navigation, and Alpamayo trajectory planning on RTX PRO 6000 and NVIDIA Thor, delivering explainable driving behavior, closed-loop lane-change scenarios, and policy validation on custom data.

MultiCoreWare Inc 10/2024 - 02/2026
Research Engineer (Embedded Systems, Sensor Engineering) | *Urbana-Champaign*

- Released VisionPilot v0.9 and v0.5, featuring a multi-threaded pipeline for sensor capture, inference, and logging/visualization in a Level 2 ADAS perception stack. Developed Closest In-Path Obstacle (CIPO) detection with obstacle motion-state estimation, and safe-distance estimation using Mobileye RSS for AEB and longitudinal control. Integrated the LaneTracker, which maintains cross-track error (CTE), yaw error, and road curvature estimates for PID-based lateral control within the system's ODD.
- Developed a middleware-agnostic, portable model deployment framework compatible with ROS2, QNX, Zenoh, and Iceoryx, using CUDA, HIP, ONNX Runtime, and TensorRT for hardware-accelerated real-time inference, including data auditing and model quantization (QAT, PTQ).
- Performed real-world ADAS testing and validation with OEMs, including vehicle demos and performance evaluation.
- Developed an empirical model for multi-sensor FMCW radar and lidar applications. Worked on optimizing sparse MIMO antenna arrays using multi-objective desirability functions (DF), balancing PSLR and beamwidth (azimuth/elevation) for grid-based designs.
- Worked on a proof of concept for llvm-mctoll targeting static binary translation between ARM and x86; focused on instruction-level debugging with GDB and validated translated binaries using Android Studio emulation.
- Worked on device enumeration in the ACPI namespace; developed Linux kernel modules; debugged I2C and UART communication.

New York University 05/2024 - 10/2024
Assistant Research Scientist | *New York*

- Developed a language-guided motion planning framework that reduced computational nodes by 27.1% and improved instruction compliance by 39.1% over baseline planners.
- Designed a multi-agent LLM architecture for high-dynamic vehicle maneuvers, iteratively refining control parameters through physics-based simulation validation and improving success rates to 90%.

TRC Robotics 10/2023 - 10/2024
Robotics Software Engineer & Business Development Associate | *Remote*

- Built and deployed Docker/Podman containers, implemented Grafana dashboards, and managed Keycloak authorization for ROS-based systems.
- Migrated servers and optimized CI/CD pipelines, improving system reliability and deployment efficiency.

PROJECTS

RL-Based Quadruped and Humanoid Control Learning | *PyTorch, MJX, Brax, TensorFlow*

- Developed and optimized PPO reinforcement learning pipelines using Brax/JAX for simulated quadruped and humanoid locomotion within the MJX physics engine.
- Built and debugged robust checkpointing inside Brax training loops, ensuring reliable model saving and resumption. Scripted end-to-end RL workflows covering environment setup, parallelized training, policy evaluation, and results visualization (video/HTML outputs).

CUDA GPU Programming Portfolio | *CUDA, C++, SIMD*

- Implemented a suite of CUDA applications (vector addition, matrix operations, convolution, softmax, image processing) with memory coalescing and shared memory tuning.
- Built SIMD-optimized CPU versions using AVX2/SSE instructions for direct performance comparison.
- Built custom CUDA conv2D kernels on image data sourced from OpenCV.

Visual Navigation Game & 3D Mapping | *OpenCV, Sensor Integration, Python, Pygame, ROS*

- Built a monocular first-person view solution with SIFT feature extraction, enabling maze exploration and image-target navigation.
- Adapted the system for real-world TurtleBot3 deployment, using IMU and camera sensors for localization and mapping.
- Data collection for 3D construction mapping: 360° cameras, 3D LiDARs, IMU, LattePanda SBC, DeepMapping2.

PUBLICATIONS

- **Weight Space Misfolding** - alphaXiv, Nov 2026
- **IWIBench: Evaluating Gradient-Free Model Driven Intra-weight Injection** - arXiv, Nov 2026
- **SimReadyBench: Evaluating models, harnesses and skills for simulation-ready asset generation** - OpenUSD benchmark, Oct 2026
- **Geometry-First Generative Worlds as Simulation Substrates in Bronchoscopy** - SurgTwin @ IROS 2026
- **Immersive Social Interaction with VR and LLM-Assisted Humanoids** - arXiv
- **Maneuver GPT-Agentic Control for Safe Autonomous Stunt Maneuvers** - IROS 2025
- **LLM-Enhanced Path Planning: Safe and Efficient Autonomous Navigation with Instructional Inputs** - arXiv
- **Explainable Autonomous Mobile Robots: Interface and Socially Aware Learning** - Late Breaking Report, IEEE RO-MAN 2024
- **A Review On Wind Farm Reliability With Hybrid Cable Connection** - Book Chapter, Energy Conversion Systems: An Overview, Nova Science Publishers. ISBN: 978-1-53619-200-1